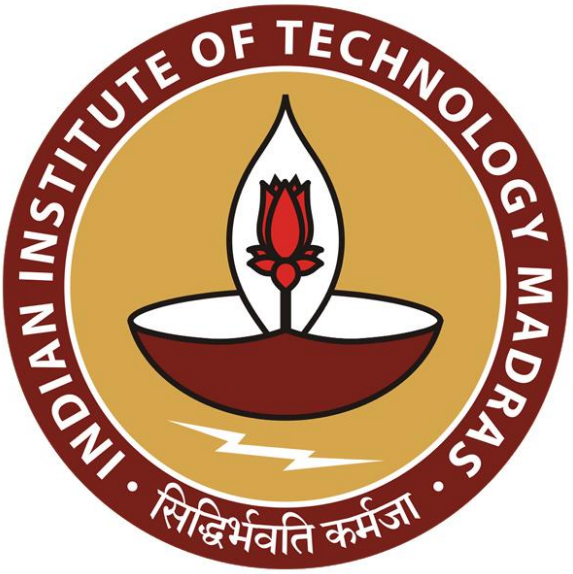
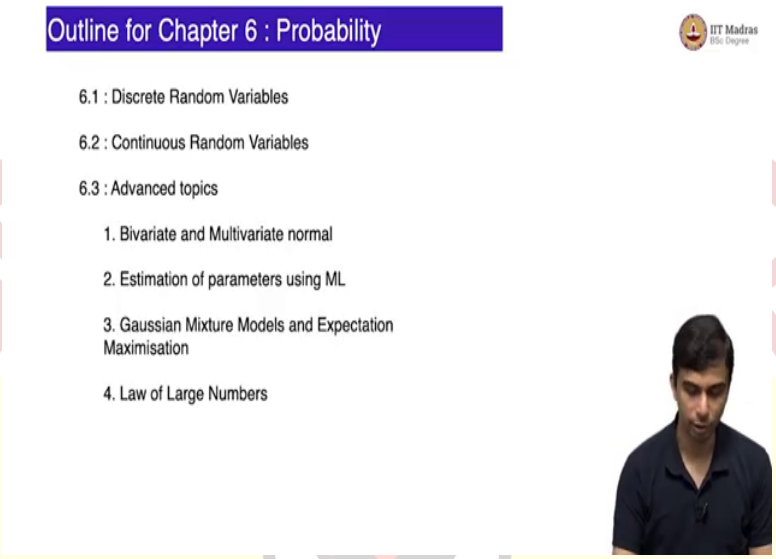


IIT Madras

ONLINE DEGREE

Machine Learning Foundations
Professor Harish Guruprasad Ramaswamy
Department of Computer Sciences and Engineering
Indian Institute of Technology, Madras
Laws of large numbers- Markov, Chebyshev, Hoeffding, Central limit

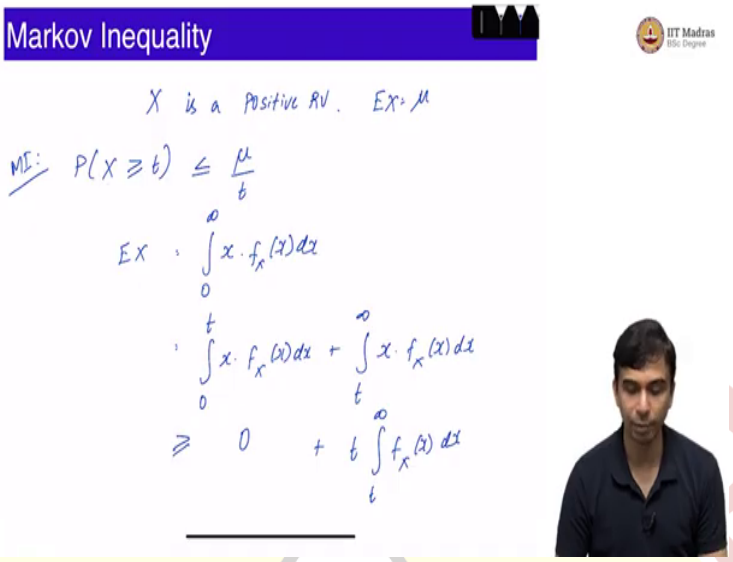
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Hello everyone and welcome to another lecture on probability. So, in this lecture we will see a collection of results which do one of two things. One, they either bound how far can this random variable be from its mean, for a given random variable how far can it be from its mean. So, recall mean was a measure of central tendency, so it was the center of mass of the density function and so on. How far can a random variable go from its mean, so that is called a tail bound.

And there is one other type of result which is for limit if you have large number of random variables and then you do a kind of average what will the effect of this average be, what will this resulting random variable be, so these are the two types of results that you will see in this video.

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So, the first of such results is the simplest and the most general which is called the Markov inequality. So, you have x is a positive random variable, we cannot take negative values let us say x is a positive random variable. So, this is what Markov inequality says, probability of x greater than or equal to some t is less than or equal to the expected value of x by t, so this is the Markov inequality statement.

So, first of all this statement would be vacuous if t was less than the mean, so if let us say x is a positive random variable and E X is equal to mu so that it is easy to write here. So, if t where less than mu then it is saying that probability of x greater than or equal to t is less than or equal to some number which is greater than 1, so that is trivially true probability is always going to be less than equal to 1.

So, this particular bound would be meaningful only for t to be greater than or equal to the mean. So, for example you can ask the question what is the probability, let us say you would happen to have a positive random variable and you can ask the question what is the and let us say the mean of that particular random variable is let say 10 meters, you have a random variable which is let us say measured in meters and the expected value of that random variable is say 10 meters, you can ask the question what is the probability that x is greater than or equal to 50 meters, you can ask the question.

And you can immediately say 50 is so that is 10 by 50 is 1 by 5, you can immediately say without saying anything the probability that this particular random variable being greater than

50 meters is less than or equal to 1 by 5 which is 20 percent, so there is only 20 percent chance that this particular random variable can exceed 50 meters.

Why is that? Because you happen to know that the mean is equal to 10. So, in fact this is a pretty tight bound, you cannot really do much better than this, this is the only information given to you, if the only information given to you about the random variable is the mean there is not much you can do this is the best that you can do and but it is a pretty simple derivation let us go ahead and do that.

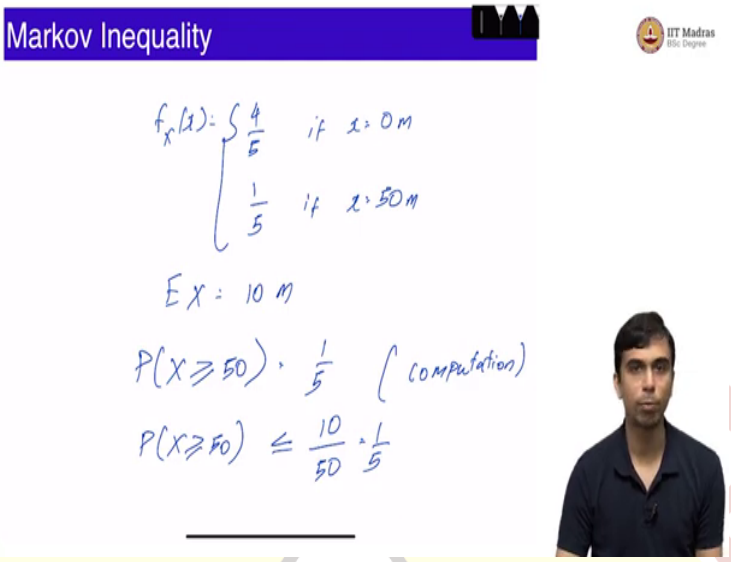
So, what is E X? E X is simply equal to integral 0 to infinity recall I am putting 0 to infinity not - infinity because x is a positive random variable, integral 0 to infinity x into f x of x dx. Now, let us break this limit into two terms just integral 0 to t into x f x of x d x plus integral t to infinity x f x of x d x.

And because your f x of x is positive and x is also positive this particular term as a whole is going to be positive. So, I can say this is greater than or equal to 0 I am going to just ignore this term and say that this term is greater than equal to 0 and what can I say about this term. So, we have x into f x of x dx.

So, the this term here x is always going to take values greater than t so I can just replace x by t and have a lower bound, so I can and this particular term becomes 0 and I am just going to put t into f x of x dx so if it is t does not depend on x, I can take it outside so t into integral t to infinity f x of x dx.

But integral t infinity of x of x dx is exactly equal to probability that x is greater than or equal to t and if you can arrange, rearrange the terms you get back this, that is probability of x greater than equal to t is less than or equal to E X by t. So, why is that? Because integral t to infinity f x of x dx is exactly equal to the probability that x is greater than or equal to t. So, this is a pretty simple proof but this serves to, serves the purpose. So, I said that this is the, this might be the best that you can do.

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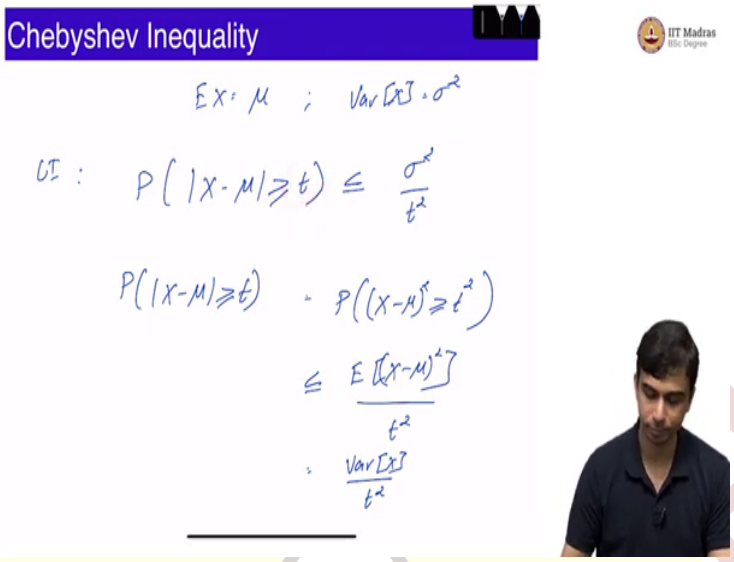
For example, let us say, let us take that example that I gave you. Let us say f x of x is equal to let us do this for a discrete random variable, this actually holds even though this under the continuous random variable section, most of these hold also for the discrete random variable in fact all of these also hold for discrete random variables.

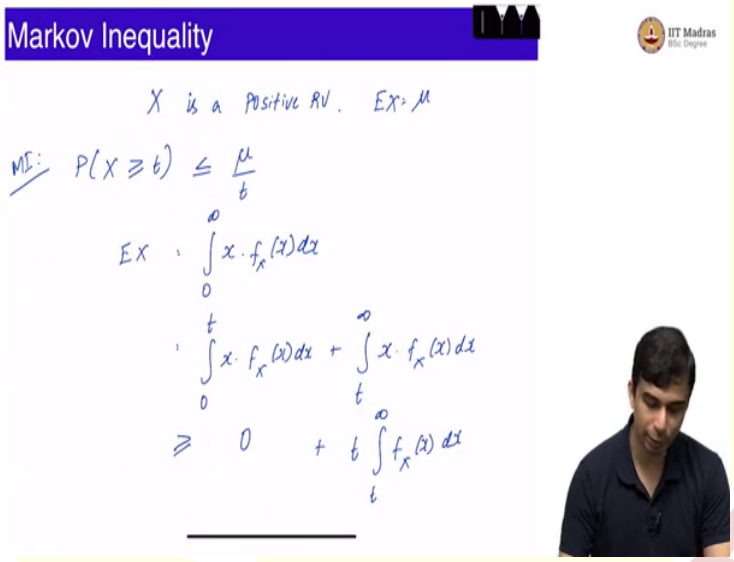
Let us say x is a discrete random variable as follows, x is equal to f x that is p m of x, the pmf of x as follows, f x of x is equal to 4 by 5 if x is equal to 0 and 1 by 5 if x is equal to 50, so that can this particular random variable takes only two values it is either 0 meters or 50 meters, 0 meters or 50 meters.

What is the expected value of this random variable? Clearly equal to 1 by 5 into 50 which is 10 meters. If you ask the question what is the probability that x is greater than or equal to 50 you know it is exactly equal to 1 by 5 this is by computation and what does Markov inequality tell you what is probability of x greater than equal to 50 that is less than or equal to expected value of x by t.

What is the expected value of x? That is 10 by 50, so in this particular case Markov inequality is actually tight. So, this is the best that you can do. Why is that? And we have given you an example where Markov inequality happens to be tight, it is not just an inequality it is actually inequality for certain cases, it is not going to be an equality for all cases but there are at least some example, some distributions where the Markov inequality is tight. So, that is an example for Markov inequality. Now, let us move on to another application.

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Let us have what is called a Chebyshev inequality. Let us say x is such that expected value of x is equal to mu and variance of x equal to sigma square, you have these two properties. Now, what does Chebyshev inequality tell you, Chebyshev inequality says probability of the absolute value of x - mu greater than or equal to some t is less than or equal to variance of let us just use sigma squared by t square.

So, both of these bounds, both Markov inequality and Chebyshev inequality are examples of what are called tail bonds, it captures how big or how far can you be from the mean. So, in the case of Markov inequality it just says probability of x is greater than or equal to t because mean was positive anywhere, so it is an x random variable. So, what is the probability that x

is very large is essentially the same as saying what is the probability that x is very far from its mean.

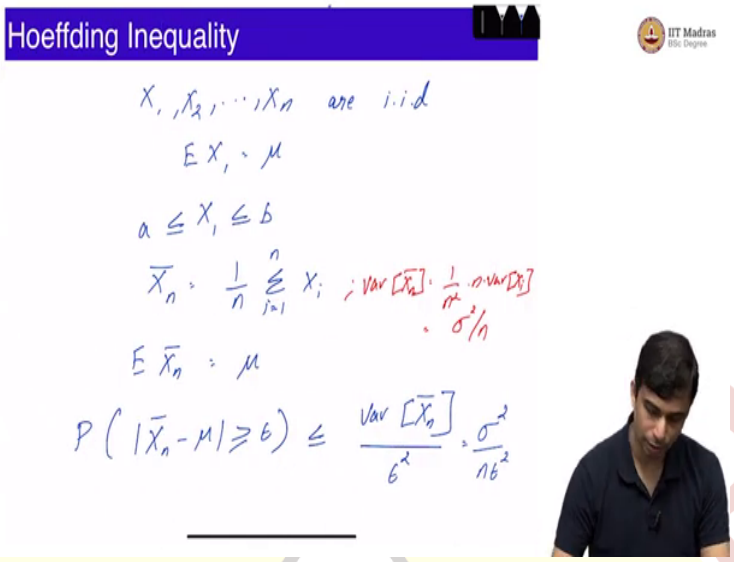
Similarly, you can have Chebyshev inequality, Chebyshev inequality is more additive in nature, so this holds even for, Chebyshev inequality also holds for, it can hold for non-positive random variables also, x can take negative values also, even in that case your Chebyshev inequality holds. This more explicitly says what is the probability that x deviates from its mean mu by more than t. What is the probability that this can happen?

And Chebyshev inequality says that is less than or equal to sigma squared by t square. So, (())(9:33) that if t is much larger than variance, let us say for example t, t squared is let us say 100 times the variance, if that is the case then sigma squared by t squared would be 1 by 100. So, if t is very large then the probability of x, x - mu greater than or equal to t is very small. But how where, how does it (()) (9:55) is 1 by t square that is the thing here. How do we prove this? The proof is quite simple, it actually follows from Markov inequality, so will do that.

Now, the left hand side is probability of absolute value of x - mu is greater than or equal to t but this is exactly equal to probability of x - mu the whole squared is greater than or equal to t square and here is where you can use the fact that x - mu the whole square is also a random variable and it happens to be positive which means that you can use your Markov inequality for this random variable x - mu the whole square, this is less than or equal to expected value of x - mu the whole square by t square.

But by now you know what this particular object is exactly equal to the variance of x. So, this is equal to variance of x by t square. So, we have that the Chebyshev inequality simply says the probability of x deviates from its mean mu by greater than or equal to some number t is less than or equal to sigma squared by t squared. So, that finishes the Chebyshev inequality.

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Then there is there are even more special inequalities which are powerful, more powerful in certain situations but it requires more conditions. Here is the (()) (11:23), let us have x several random variables X 1, X 2, dot dot X n you have these n random variables which are identically distributed and independent.

So, you will often see this written in short hand as X 1 to X n are i.i.d, so i.i.d is short form for independent and identically distributed. So, X 1 to X n are i.i.d, let us say expected value of X i is equal to mu, it does not matter what i is, they are i.i.d. So, expected value of X i is equal to expected value of X j, whatever, expected value of x.

Let us make it simple, the expected value of X 1 is equal to mu and let us say X 1 is between, bounded between a and b, so let us change the direction X 1 is between a and b. So, these random variables are independent and identically distributed have mean mu and are bounded between a and b.

For example, X 1 to X n could be uniform a, b or they could be anything just that they should be bounded between a and b. If this is the case let us define another random variable now X n bar, X n bar is simply equal to 1 by n sigma i equal to 1 to n X i, so X n bar is simply the empirical average the sample mean of these n random variables.

So, you already have a guess that X n bar should be pretty close to mu. Why is that? Well, what is expected value of X n bar? X n bar is also a random variable, X n bar is a function of the for n random variables X 1 to X n so it is also a random variable, expected value of X n

bar by linearity of expectation is going to be equal to mu. Why is that? Because 1 by n sum over i equal to 1 to n mu which is equal to mu.

Now, we can ask the question, what is the probability that X n bar and mu are far apart? Greater than or equal to some epsilon? You can ask the question what is the probability that X n bar and mu are far apart? Immediately you can apply your Chebyshev inequality. Why is that? Because X n bar is a random variable, you can apply that and get some answer, let us do that first.

So, if you do that what is the probability of X n bar - mu greater than equal to epsilon that is simply less than or equal to the variance of X n bar by epsilon square. What can you say about variance of X n bar? X n bar is 1 by n times sum over i equal to 1 to n X i which means that variance of X n bar is equal to 1 by n squared into sum of the variance of these things because X 1 to X n are independent so 1 by n squared into n into variance of X i so which is equal to sigma squared by n.

So, substituting that here you get this is equal to sigma squared by n epsilon square, this is a perfectly valid bound, I am not saying this is invalid you can use that so we have that probability of X n bar - mu is greater than equal to epsilon is less than or equal to sigma squared by n epsilon squared, which is perfectly fine it is a valid statement.

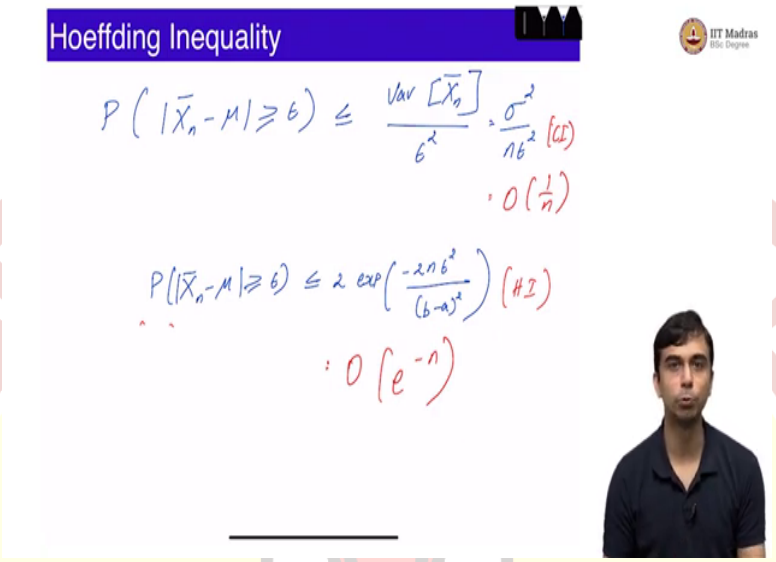
But what you are not actually exploiting the full power of the mean. So, the mean, the sampled mean concentrates quickly on to its the true, the true mean much faster than 1 by n. So, what this particular thing says is probability of X n bar - mu greater than or equal to epsilon goes down as 1 by n that is what this is saying.

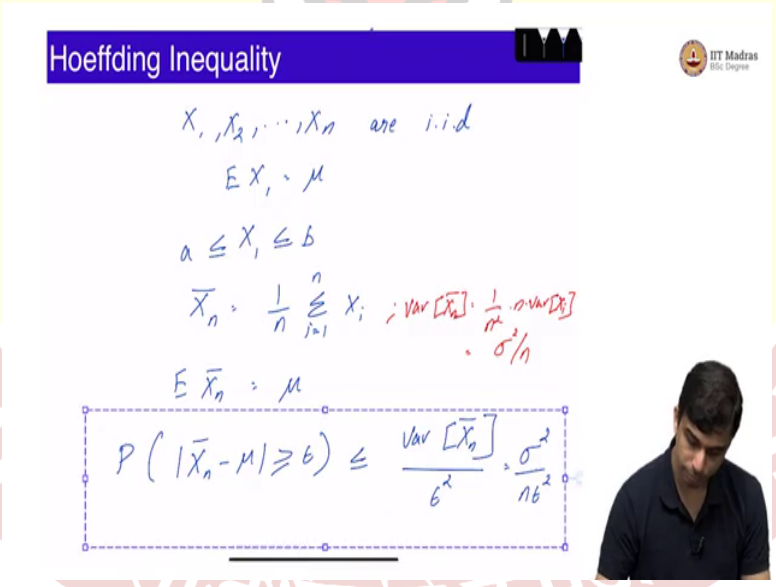
So, as n goes from let us say 10 to 100 this probability of X n bar - mu greater than or equal to epsilon goes from 1 by 10 to 1 by 100 let us ignore the sigma square and epsilon square terms here, just you increase the number of points in number of random variables from 10 to 100 if that happens then this the probability of the deviation goes down from 1 by 10 to 1 by 100.

But that is not, that is what this bond is saying but in reality that is not the case. In reality it goes down much more than that. And why is that the case? Because Chebyshev inequality only uses one property of the sample mean, the sample, if that the variance of X n bar is equal to sigma squared by n.

So, the sample mean has other properties than just the variance and there is an inequality which is called the Hoeffding inequality which exploits this in a much better way and you get this bound.

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So, I am not going to prove this bound here, but I am going to write that down though and this is as follows the Hoeffding inequality says that probability of absolute value of X n bar - mu greater than or equal to sum epsilon is less than or equal to 2 times exp - 2 n epsilon squared by b - a the whole square.

So, what is the difference between both of, both the Chebyshev and this is Markov, so this is Hoeffding inequality, if you did the Chebyshev inequality this is what you have got, let us copy paste that. In both cases the left hand side is the same, so both Hoeffding inequality and

Chebyshev inequality the left hand side is the same, but the right hand side is different, the main thing that you want to look at is that, is the how does the right hand side gets small as n gets bigger.

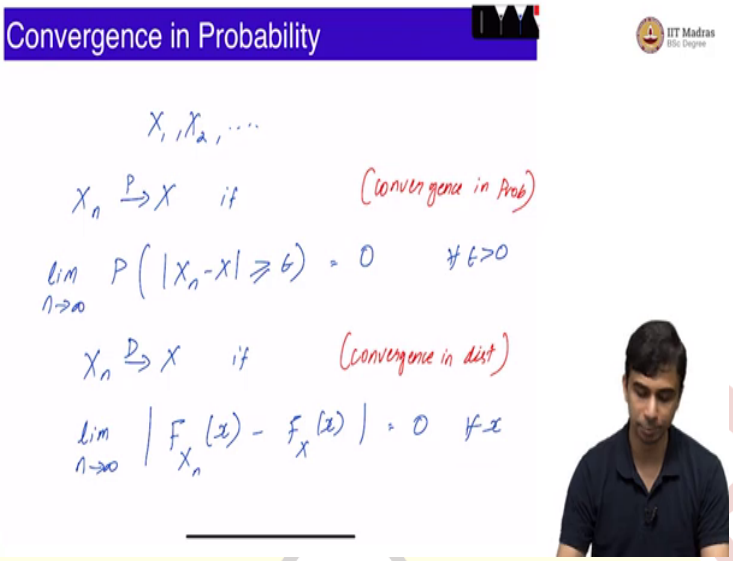
In the Chebyshev inequality it goes down as 1 by n, in the Hoeffding inequality it goes down as e power - m, so these other terms are constant they do not depend on n, so there is only one term here, so this goes exponentially down with n this goes only reciprocal 1 by n. So, in this case this is O of 1 by n you might just see this is written as O of 1 by n this is O of e power - n.

So, if you do not understand the O of notation, do not worry about this. What this is broadly saying is you can ignore the other factors sigma square, epsilon square and so on. If you do that it is O of 1 by n and this on your hand if you ignore the other factors it is e power - n. So, e power - n goes to 0 with increasing n much faster than 1 by n.

Both of these e power - n and 1 by n both of these go to 0 as n goes to infinity. But e power - n is goes to 0 much faster than 1 by n. So, that is why Hoeffding inequality is better for this particular special case, Chebyshev inequality is more general, Chebyshev inequality holds for general random variables, we just need the variance. But Hoeffding inequality holds only for the sampled mean.

There are some generalizations of the sample mean but broadly you will mostly see the Hoeffding inequality applied only for the sample mean. So, we have not proved this because not it is not important right now you only need to understand that the sample mean concentrates around the true mean at the rate of e power - n, so that is the, this is called as concentration inequality, because as you increase your number of random variables, the sample mean will almost all will be very, very close to the true mean. Yes, okay, so with that we will move to the next concept, which is convergence in probability.

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Let us say we have a sequence of random variables; we have a sequence of random variables. So, we have a sequence of real numbers, we have defined the convergence of a sequence of real numbers. How do we do that? Well, we say this sequence of real numbers converges to another real number if we can draw a circle around mean, we can this particular sequence approaches this real number.

So, for example we might say it eventually reaches within an epsilon ball of the limit and stays within that epsilon ball, that was the definition of the convergence of a sequence. But now we have a sequence of random variables X 1, X 2 and so on. So, random variables are not clearly not real numbers, so they are more than real numbers, they are actually functions from omega to r. So, can we have notions of convergence?

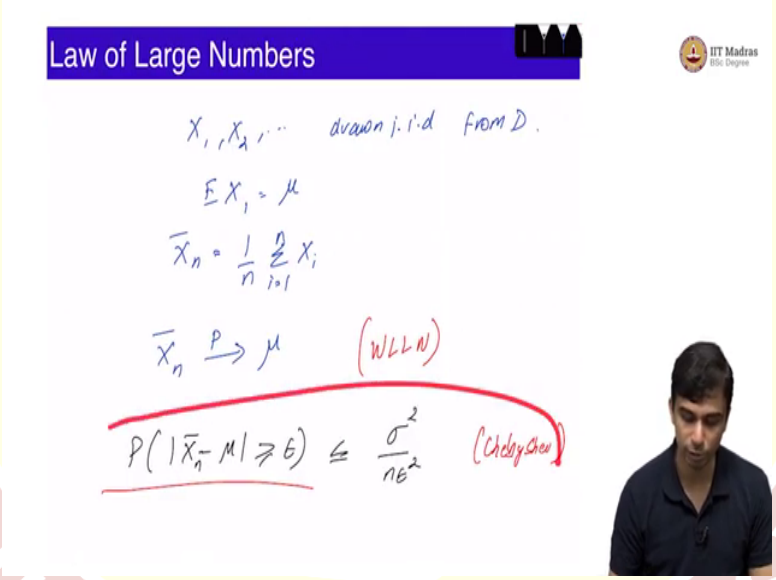
There are several notions that you can do, the most popular one is, the most natural one is what is called the convergence in probability. We say that X n converges to X in probability if the probability limit n approaches infinity the probability of absolute value of X n - x greater than or equal to epsilon is equal to 0 for all epsilon greater than 0.

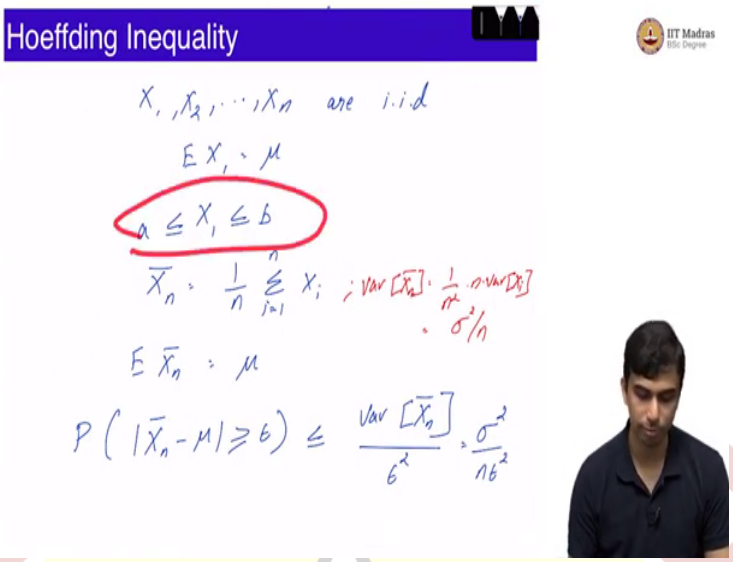
So, you can fix any arbitrarily small epsilon as long as it is not 0 and you just measure the probability that absolute value of X n - X is greater than equal to epsilon and this particular probability should go to 0 as n goes to infinity, so this is called convergence in probability. There are other notions of convergence such as convergence almost surely which we will not do so that is a stronger condition, so but there is another weaker condition which is simply called convergence in distribution.

We will say X n, so this is convergence in probability, there is another notion of convergence called convergence in distribution we will say X n converges to X in distribution d if limit n approaches infinity your cdf function F X n of x - f x of x equal to 0 for all x. So, whatever be the x your cdf of the random variable X n and the cdf of the random variable X are matching so this is a much weaker notion, so convergence in probability implies convergence in distribution but not the other way around.

There is an even stronger notion of than convergence in probability which is called convergence almost surely but we will not do that, we will only mostly deal with these two notions of convergence, this is convergence in distribution, there are other notions of convergence also which you might do if you are doing a course in probability theory. So, why are we doing this?

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So, with this we are, we have the machinery to define two fundamental theorems of probability. First is the law of large numbers. What does the law of large numbers say? Law of large number simply says as follows that is you have X 1, you have let us say X 1, X 2, dot dot dot drawn i.i.d from some distribution D. And you have expected value of X 1 is equal to mu, this is the case what the law of large numbers says is as follows.

It is X define X n bar as before is 1 by n sigma i equal to 1 to n X i. What the weak law of large number says is X n bar converges in probability to the constant mu. So, this is what the weak law of large numbers states, this is called the weak law of large numbers, you might get, might ask what is the strong law of large numbers, strong large numbers simply says that X n converges to mu almost surely but will not be worried about that, we will just worried about the weak law of large numbers.

The weak law of large number says that X n bar converges in probability to the constant mu. How do we prove that? Well, the proof for this for with an extra assumption is very simple, you assume that the variance exists. What does that mean? That means that the variance is not infinity.

In sigma you assume that each of these random variables X i also have a variance and that happens to be equal to sigma squared which is not infinity, if that is the case then simply by Chebyshev inequality we have that X n bar is probability of X n bar - mu, absolute value of X n bar - mu greater than or equal to epsilon is less than or equal to sigma squared by n epsilon square, this is by Chebyshev inequality.

So, if n goes to infinity this term clearly goes to 0 so probability of X n bar - mu greater than equal to epsilon goes to 0 whatever be the value of epsilon, whatever be the fixed value of epsilon as n goes to infinity this goes to 0, so we have that the limit we have this particular condition is satisfied, limit n approaches infinity probability of X n bar - x greater than or equal to epsilon is indeed equal to 0.

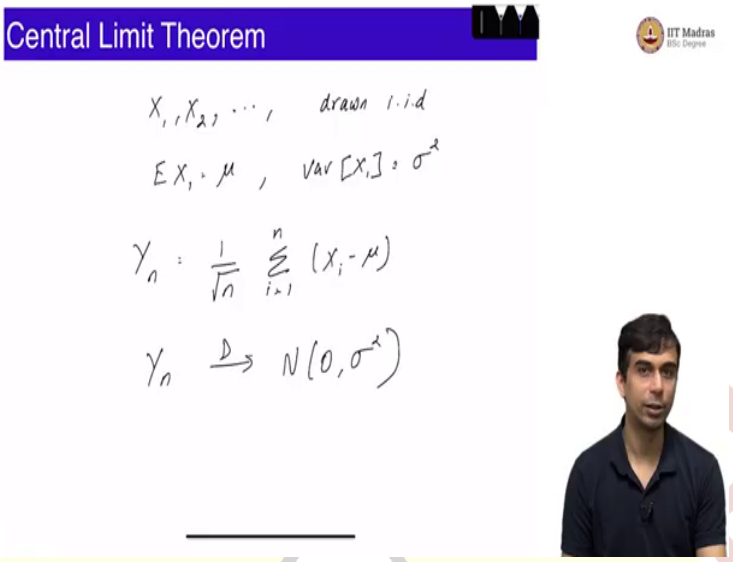
So, of course it seems like we are making an extra assumption that is sigma squared is not infinity, sigma squared can be infinity for certain distribution sigma squared can be infinity. But let us what we can prove weak law of large numbers very easily if you assume that the variance is not infinity, you can actually prove that even without that but that is a much more complicated proof.

We can ask the question why did not we use Hoeffding inequality for this, you can immediately guess why, for Hoeffding inequality have to make a much stronger assumption that is each of these random variables are bounded. Recall for Hoeffding inequality you have to have this extra thing which is that each of these random variables are bounded between a to b.

If that is the case, you can indeed use Hoeffding inequality but saying the random variables are bounded is much more stronger than saying that the variance is not infinity. So, if the random variable is bounded clearly the variance is going to be finite. But it is not the other way around, the variance can be infinity, sorry variance can be finite but the random variable can still be unbounded, the standard example is the normal distribution.

So, for example you had the normal distribution where you can see that the variance is finite it is let us say standard normal the variance is equal to 1, but it the random variable is clearly not bounded, so e power - z squared by 2 is not 0 for any z so it is not bounded, that is the reason why we did not use Hoeffding inequality but rather we use Chebyshev inequality to prove the weak law of large numbers, still we are making the assumption that the variance is finite. So, that is the weak law of large numbers.

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So, now we will move on to another fundamental theorem of probability which is the central limit theorem. The setting is similar to the case of the law of large numbers where you have X 1, X 2, dot dot dot several of these random variables drawn i.i.d. So, let us have expected value of X 1 is mu and variance of X 1 is equal to sigma square.

Now, we will define another related random variable which is very similar to the sample mean but subtly different, we will define Y n as 1 by square root of n sum over i equal to 1 to n X i - mu. This looks very similar to the sample mean you have sum over i equal to 1 to n, there is you have X i - mu here that is anyway we can imagine mu is 0 so that this term is sum over i equal to n X i.

But the main difference is you have 1 by square root of n and not 1 by n so that is the difference between Y n and the sample mean. What the central limit theorem says is as follows. It says Y n converges in distribution D, converges distribution to the normal distribution with mean 0 and variance sigma square.

Note that we are not saying its converges in probability but rather converges in distribution and the right hand side is not a constant but rather a distribution for law of large numbers weak law of large numbers we had a similar object as Y n which is X n bar on the left side and the right side you had a constant. So, law of large numbers all three of these things were different, you had X n bar there, you had convergence in probability there and you had a constant.

Here you have Y n which is not the same as X n bar but slightly different, you have convergence in distribution and most importantly in the right side you do not have a constant but rather a distribution, a density. So, what does this mean? So, we have on the right side we have a normal distribution.

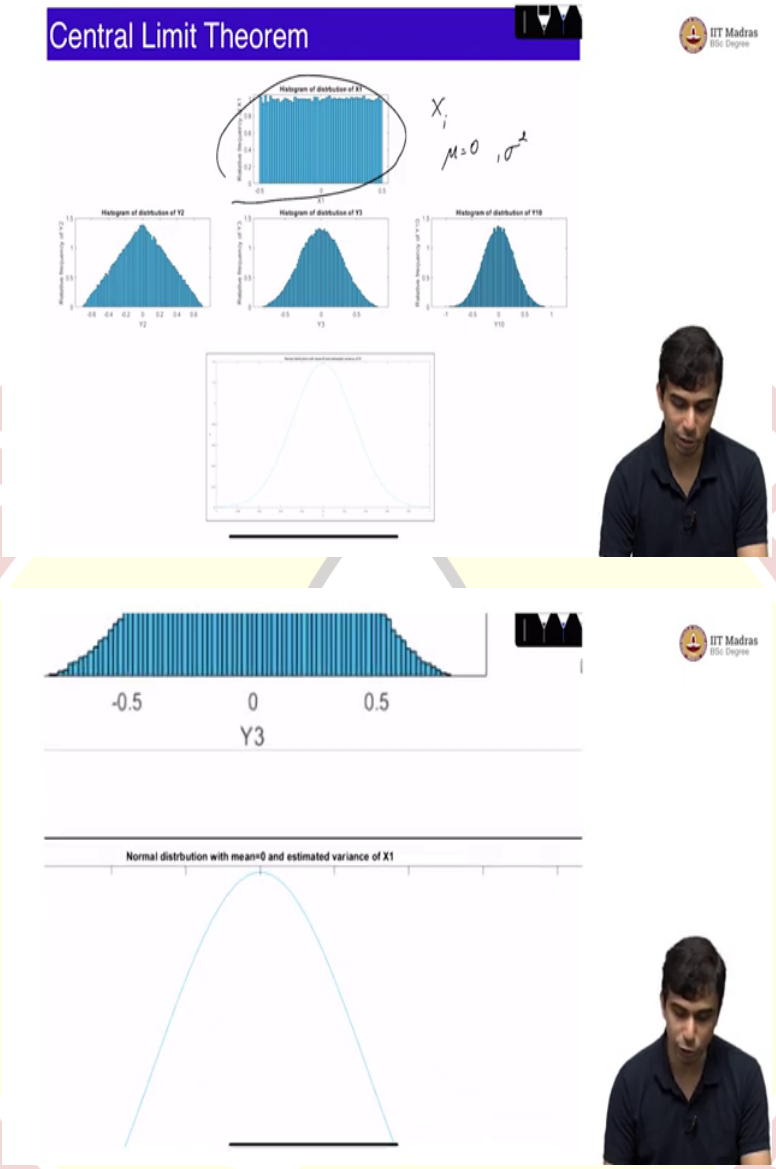
You can view Y n as essentially some scaled sum of random variables. So, you have a scaled sum of random variables, if you have a scaled sum of random variables you get a normal distribution. So, this is the reason why you the normal random variable is ubiquitous in nature. So, for example I told you that if you measured the current of just a piece of wire using a very sensitive ammeter what you will get is a normally distributed random variable.

Why is that? Because the current is effectively the result of summation of several electrons moving in one direction. So, you are simply adding the number of electrons that move from one side. So, the current that you are measuring is just an addition of several random variables, they all happen to be independent and because of that you get a natural center limit theorem essentially applies there and you have the noise that is measured in the ammeter is going to be normally distributed.

And this is going to be true for almost any practical situation where you, the thing that you are measuring has an additive effect of several independent, each of these independent things can be in any distribution. For example, in the case of electron it was just Bernoulli, it is going to left to right or right to left and so on, so that was just a Bernoulli random variable but you add on, you add several of these random variables and you average them, what you get is a scaled normal distribution.

So, this is going to be a very powerful theorem and in fact even though it looks like it only it is a limit statement, it only holds for large enough n but in fact central limit theorem begins to hold even for n in the range of 10 to 20, even in the range of n equal 10 to 20 the central limit theorem begins to you can see the effect of central limit theorem.

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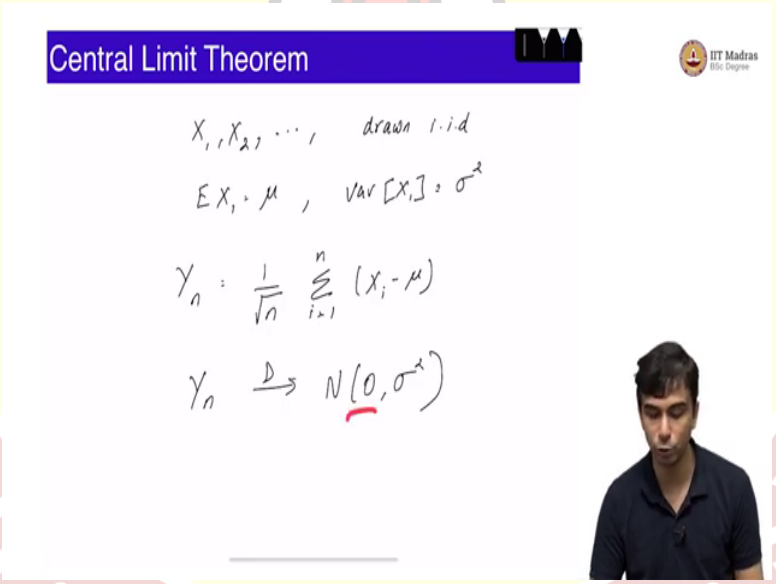
So, here is a very simple example. Here you have your X i's are distributed uniformly in the range of - 0.5 to 0.5. So, in this case mu happens to be equal to 0, sigma squared you can compute, it can be something, sigma squared is not equal to 1 but it is something, it does not matter what it is. So, you have these X i's are such that mean is 0 and sigma squared is something, the variance is something.

So, you have these X i's, so you can compute the, this is the histogram of the X i's, you essentially sample so many X i's and you measure how many times. So, histogram simply counts, simply gives the counts. So, how many times is in the range of 0 to 0.01, how many times is in the range of 0.01 to 0.02 and so on so that is the histogram. So, that is the, you can see it visualized in a bar plot here.

We can do the same thing, we can compute Y 2, Y 3, Y 10 and so on. And you can compute the histogram of these random variables also and you can see the shape of the histogram, you can see a triangle shape coming here you can see a piecewise quadratic of sorts coming here and then for Y 10 you get a shape which looks extremely similar to the density of the bell shaped curve of the normal plane.

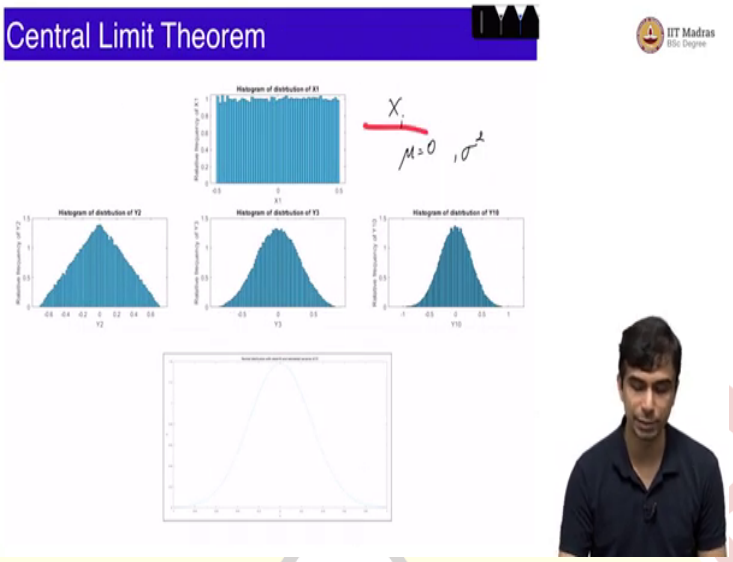
In fact, we can compute that, also we can compute the normal, you can compute the normal distribution with mean 0 and variance given by the variance of X 1 and the curve looks like this. In fact, you can overlay this on top of the histogram of Y 10 and it would almost perfectly match. So, what this says is Y 10 already the density of Y 10, Y 10 is already the distribution of Y n is already pretty close to the normal with mean given by the mean of X 1, sorry the mean given by 0 and variance given by variance of X 1. So, that is the center limit theorem.

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Here is the statement again, so Y n is defined as 1 by square root of n sum over i equal to 1 to n X i - mu which means that Y n is distributed as if that is the case then Y n is converges in distribution to the normal with mean 0 and variance sigma square.

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And here is an illustration of that for a very simple X i where they are all i.i.d uniform from some range. So, with that we will wrap up the central limit theorem and the final module of this particular chapter, this week.